

Representations of 9^k as a Sum of Three Positive Squares: a Proof of Seidov's Conjecture on OEIS A047926

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Abstract

Let $A047926(n) = (3^{n+1} + 2n + 1)/4$. In March 2012, Z. Seidov conjectured that this sequence counts the representations of 3^{2n} as a sum $a^2 + b^2 + c^2$ with $0 < a \leq b \leq c$. We prove the conjecture, after correcting an off-by-one in its stated indexing: for every $k \geq 1$ the number of such representations of 9^k is $(3^k + 2k - 1)/4 = A047926(k - 1)$. We prove moreover the finer statement that exactly k of these representations have two equal parts and $(3^k - 2k - 1)/4$ have three distinct parts, that none has three equal parts, and that 9^k is never a sum of two positive squares. The argument deflates the classical count $r_3(9^k) = 6(2 \cdot 3^k - 1)$, due to Pall and reproved elementarily by Hirschhorn and Sellers, to unordered triples of positive integers; the two degenerate shapes are controlled by the divisor counts attached to the forms $x^2 + y^2$ and $x^2 + 2y^2$, both of class number one.

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1 Introduction

The sequence A047926 of the On-Line Encyclopedia of Integer Sequences [5] is defined by the closed form

$$a(n) = \frac{3^{n+1} + 2n + 1}{4}, \quad n \geq 0,$$

with initial terms

$$1, 3, 8, 22, 63, 185, 550, 1644, 4925, 14767, 44292, 132866, \dots$$

Its original interpretation is combinatorial: it is the density of a certain regular language over a four-letter alphabet, contributed by N. Moreira and R. Reis [2]. On 1 March 2012, Zak Seidov contributed to the entry the following comment:

Conjecture: Number of representations of $3^{(2n)}$ as a sum $a^2 + b^2 + c^2$ with $0 < a \leq b \leq c$. That is, $a(1) = 3$ because $3^2 = 1^2 + 2^2 + 2^2$, $a(2) = 3$ because $3^4 = 1^2 + 4^2 + 8^2 = 3^2 + 6^2 + 6^2 = 4^2 + 4^2 + 7^2$.

Throughout we write

$$P(N) = \#\{(a, b, c) \in \mathbb{Z}^3 : 0 < a \leq b \leq c, a^2 + b^2 + c^2 = N\},$$

and we write $r_3(N)$ for the number of $(x, y, z) \in \mathbb{Z}^3$ with $x^2 + y^2 + z^2 = N$, counting signs and order.

The indexing

The comment as punctuated is inconsistent with itself, and it is worth recording at once which reading is intended. Taken literally it asserts $P(3^{2n}) = a(n)$; but $3^2 = 9$ admits the single representation $1^2 + 2^2 + 2^2$, so $P(3^2) = 1$, whereas $a(1) = 3$. The two examples supplied resolve the matter: they exhibit one representation of 3^2 and three of 3^4 , that is, the first two terms 1, 3 of the listed data, so the count for 3^{2n} is the n -th term of the *list* rather than the term of index n . Since the offset of A047926 is 0, the intended assertion is

$$P(3^{2n}) = a(n-1), \quad n \geq 1.$$

Under the literal reading the statement is false at every index; under the shifted reading it is true, and that is what we prove. (The label “ $a(1) = 3$ ” attached to the example for 3^2 appears to be a slip for “ $a(0) = 1$ ”.)

Theorem 1. *For every integer $k \geq 1$,*

$$P(9^k) = \frac{3^k + 2k - 1}{4} = a(k-1).$$

The right-hand side is an integer: if k is odd then $3^k + 2k - 1 \equiv 3 + 2 - 1 \equiv 0 \pmod{4}$, and if k is even then $3^k + 2k - 1 \equiv 1 + 0 - 1 \equiv 0 \pmod{4}$.

We prove Theorem 1 in the sharper form of Corollary 2, which resolves the count into its two shapes and so settles the question whichever way one scopes it.

Corollary 2. *Let $k \geq 1$. Among the representations $9^k = a^2 + b^2 + c^2$ with $0 < a \leq b \leq c$, exactly k satisfy $a = b$ or $b = c$, and exactly $(3^k - 2k - 1)/4$ have $a < b < c$. None satisfies $a = b = c$, and 9^k is not a sum of two positive squares for any k .*

2 The classical input

The one non-elementary ingredient is a classical evaluation of r_3 along powers of 9. We state it in the form we need and record two independent derivations.

Lemma 3. *For all $k \geq 0$,*

$$r_3(9^k) = 6(2 \cdot 3^k - 1) = 12 \cdot 3^k - 6.$$

Proof. Hirschhorn and Sellers [1], writing s for r_3 , prove by elementary q -series methods (their equation (1), established for $\lambda = 1$ in their §4 and completed by induction in their §5) that for $\lambda \geq 1$

$$s(3^{2\lambda}(3n+1)) = (2 \cdot 3^\lambda - 1) s(3n+1).$$

Take $n = 0$, so that $3n+1 = 1 \equiv 1 \pmod{3}$. This gives $r_3(9^\lambda) = (2 \cdot 3^\lambda - 1) r_3(1)$, and $r_3(1) = 6$. The case $k = 0$ reads $6 = 6$. $\square$

Remark 4. The underlying relation $s(p^2n) = ((p+1) - \left(\frac{-n}{p}\right))s(n) - p s(n/p^2)$ for odd primes p is due to G. Pall [3], who derived it from the arithmetic of quaternions; Hirschhorn and Sellers rediscovered the $p = 3$ case and record Pall's priority.

Remark 5 (Independent derivation). Lemma 3 may also be read off from Gauss's theory. Writing r_3^* for the number of *primitive* representations, one has $r_3(N) = \sum_{d^2|N} r_3^*(N/d^2)$ and, for $m \equiv 1, 2 \pmod{4}$ with $m > 4$, $r_3^*(m) = 12 h(-4m)$, where $h(-4m)$ is the class number of the order of discriminant $-4m$. For $m = 9^i$ that order is the order of conductor $f = 3^i$ in $\mathbb{Q}(i)$, whose class number is

$$h = \frac{h_{\mathbb{Q}(i)} f}{[\mathcal{O}^\times : \mathcal{O}_f^\times]} \prod_{p|f} \left(1 - \frac{(-4/p)}{p}\right) = \frac{3^i}{2} \left(1 + \frac{1}{3}\right) = 2 \cdot 3^{i-1},$$

since $h_{\mathbb{Q}(i)} = 1$, the unit index is 2, and $(-4/3) = -1$. Hence $r_3^*(9^i) = 8 \cdot 3^i$ for $i \geq 1$, while $r_3^*(1) = 6$, and

$$r_3(9^k) = 6 + \sum_{i=1}^k 8 \cdot 3^i = 12 \cdot 3^k - 6.$$

Both the aggregate and the primitive counts were confirmed numerically; see §6.

3 The degenerate shapes

Three lemmas isolate the representations that are not triples of distinct positive integers.

Lemma 6. *For every $k \geq 1$ there is no pair of integers $b, c > 0$ with $b^2 + c^2 = 9^k$. Consequently the only representation of 9^k with a zero coordinate is $(0, 0, \pm 3^k)$ up to order.*

Proof. By Jacobi's two-square theorem, $r_2(n) = 4(d_1(n) - d_3(n))$, where $d_j(n)$ counts the divisors of n congruent to j modulo 4. The divisors of 3^{2k} are the powers 3^j with $0 \leq j \leq 2k$, and $3^j \equiv 1 \pmod{4}$ exactly when j is even. Hence $d_1 = k+1$, $d_3 = k$, and $r_2(3^{2k}) = 4$. The four pairs $(\pm 3^k, 0)$ and $(0, \pm 3^k)$ already account for this count, so no representation has both coordinates nonzero. $\square$

Lemma 7. *For every $k \geq 1$ the equation $3a^2 = 9^k$ has no integer solution.*

Proof. It would force $a^2 = 3^{2k-1}$ with $2k-1$ odd, impossible by unique factorisation. $\square$

Lemma 8. For every $k \geq 1$,

$$E_k := \#\{(a, c) \in \mathbb{Z}_{>0}^2 : 2a^2 + c^2 = 9^k\} = k.$$

Proof. The form $x^2 + 2y^2$ has discriminant -8 and class number one, and the associated order has exactly two units, so for $n > 0$ the number of representations is

$$R(n) = 2 \sum_{d|n} \chi_{-8}(d), \quad \chi_{-8}(d) = \begin{cases} +1, & d \equiv 1, 3 \pmod{8}, \\ -1, & d \equiv 5, 7 \pmod{8}, \\ 0, & d \text{ even.} \end{cases}$$

The divisors of 3^{2k} are the $2k + 1$ powers 3^j , and $3^j \equiv 1 \pmod{8}$ for j even, $3^j \equiv 3 \pmod{8}$ for j odd; in either case $\chi_{-8}(3^j) = +1$. Hence $R(9^k) = 2(2k + 1) = 4k + 2$.

Among these, the solutions with $y = 0$ are $x = \pm 3^k$, two in number, and there is none with $x = 0$, since $2y^2 = 3^{2k}$ is insoluble. The remaining $4k$ solutions have $x \neq 0$ and $y \neq 0$ and fall into orbits of size four under the two sign changes, so exactly k of them have $x, y > 0$. By Lemma 7 none of these has $a = c$, so each determines a distinct multiset $\{a, a, c\}$ with $a \neq c$. $\square$

4 Proof of the Theorem

Proof of Theorem 1 and Corollary 2. Fix $k \geq 1$ and put $N = 9^k$. Partition the $r_3(N)$ signed ordered triples (x, y, z) with $x^2 + y^2 + z^2 = N$ according to the multiset $\{|x|, |y|, |z|\}$. Each multiset contributes (number of distinct orderings) $\times$ (number of admissible sign patterns), a zero entry admitting one sign and a nonzero entry two.

By Lemma 6 the shape $\{0, b, c\}$ with $b, c > 0$ does not occur, and by Lemma 7 neither does $\{a, a, a\}$ with $a > 0$. Exactly three shapes remain:

shape of $\{ x , y , z \}$	orderings	sign patterns	weight
$\{0, 0, 3^k\}$	3	2	6
$\{a, b, c\}, 0 < a < b < c$	6	8	48
$\{a, a, c\}, a, c > 0, a \neq c$	3	8	24

Let D_k be the number of multisets of the second shape and E_k the number of the third. Then

$$r_3(9^k) = 6 + 48 D_k + 24 E_k.$$

By Lemma 3 the left-hand side equals $12 \cdot 3^k - 6$, and by Lemma 8 we have $E_k = k$. Hence

$$48 D_k = 12 \cdot 3^k - 12 - 24k, \quad \text{that is} \quad D_k = \frac{3^k - 2k - 1}{4},$$

which is a nonnegative integer since $3^k \geq 2k + 1$ for $k \geq 1$. Finally

$$P(9^k) = D_k + E_k = \frac{3^k - 2k - 1}{4} + k = \frac{3^k + 2k - 1}{4} = a(k - 1). \quad \square$$

5 Where the hypothesis is used

The base 3 and the even exponent are not decorative; each of the three steps uses a distinct arithmetic property of 3, and each would fail for a general modulus.

- $9^k \equiv 1 \pmod{3}$ places it in the *first* of the four cases of Hirschhorn and Sellers, the one with multiplier $2 \cdot 3^\lambda - 1$; their other cases carry the multipliers 3^λ and $(3^{\lambda+1} - 1)/2$, which would give a different leading term.
- $3 \equiv 3 \pmod{4}$ forces $r_2(3^{2k}) = 4$ in Lemma 6, killing the shape $\{0, b, c\}$ outright. For a base congruent to 1 modulo 4 that shape is present and contributes.
- $3^j \in \{1, 3\} \pmod{8}$ for every $j \geq 0$ makes every divisor count +1 in Lemma 8. This is the sole source of the linear term $2k$ in the final answer; without it E_k would be bounded rather than growing.

The same three-step recipe — a classical multiplier for r_3 , a two-square count, and a count by $x^2 + 2y^2$ — evaluates $P(N)$ for any N whose r_3 is known. In that sense the conjecture is an instance of a general procedure rather than an isolated identity, and this is the reason it admits a short proof.

Proposition 9. *For $k \geq 1$ the proportion of representations with two equal parts tends to zero: $E_k/P(9^k) = 4k/(3^k + 2k - 1) \rightarrow 0$. The first values of $(P(9^k))_{k \geq 1}$ are 1, 3, 8, 22, 63, 185, 550, 1644, 4925, of which 1, 2, 3, 4, 5, 6, 7, 8, 9 respectively have two equal parts.*

Proof. Immediate from Theorem 1 and Lemma 8; the listed values are the outputs recorded in §6. $\square$

6 Computational verification

The results above are proved, so computation serves only as a guard against error in their statement. A single script (Python 3, standard library only) performs the following checks; all pass.

1. The closed form $(3^{n+1} + 2n + 1)/4$ reproduces all 27 terms in the published DATA field of A047926.
2. Brute-force enumeration of $\{0 < a \leq b \leq c : a^2 + b^2 + c^2 = 9^k\}$ for $k = 1, \dots, 9$ returns 1, 3, 8, 22, 63, 185, 550, 1644, 4925, matching $(3^k + 2k - 1)/4$ in every case. The largest instance is $9^9 = 387\,420\,489$.
3. Direct evaluation of $r_3(9^k)$ for $k = 0, \dots, 7$ returns 6, 30, 102, 318, 966, 2910, 8742, 26238, matching $12 \cdot 3^k - 6$.
4. Direct evaluation of the primitive counts $r_3^*(9^k)$ for $k = 0, \dots, 5$ returns 6, 24, 72, 216, 648, 1944, matching Remark 5.

5. For $k = 1, \dots, 12$, the number of pairs $0 < b \leq c$ with $b^2 + c^2 = 9^k$ is 0, as asserted in Lemma 6.
6. For $k = 1, \dots, 13$, the number of pairs $a, c > 0$ with $2a^2 + c^2 = 9^k$ equals k , as asserted in Lemma 8.
7. For $k = 1, \dots, 40$, the equation $3a^2 = 9^k$ has no solution, as asserted in Lemma 7.

Every numeral quoted in this note is either an output of that script or a figure taken from a cited source.

7 Concluding remarks

We make no claim as to the difficulty of the result. Its substance is a bookkeeping step: once $r_3(9^k)$ is known, the passage to unordered triples of positive integers requires only that one control the two degenerate shapes, and both are governed by class-number-one forms whose representation counts are divisor sums. The classical evaluation of $r_3(9^k)$ is due to Pall [3] and, in the elementary form quoted here, to Hirschhorn and Sellers [1]; nothing in §2 is new.

Two caveats are recorded for the reader's benefit. First, the correction of the indexing in §1 is essential: as literally written the conjecture is false, and a proof addressed to the literal statement would be a disproof. Second, every copy of the OEIS entry the author was able to retrieve is a cached snapshot dated 19 January 2019, the search and history endpoints of the OEIS being closed to automated access; the conjecture was certainly still open then, and no proof of it was found in the literature, but the author has not been able to verify directly that no comment settling it has been added since. Prior knowledge of so short an argument would not be surprising, and the author would welcome notice of any earlier appearance.

References

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